

Ethics

Welcome back to CS 2100!

Prof. Rasika Bhalerao

Ethics Lecture learning outcomes:

- Identify pros and cons of using given data to solve the desired problem
- Identify and defend privacy issues
 - Eventually, do this without being told there is a privacy issue to identify
- Identify stakeholders, and values and interests at stake for each stakeholder
 - Identify and discuss conflicts of values
 - Identify modifications that would mitigate conflicts of values

NOT learning outcomes:

- Tell students what to think / which opinions they should hold

Khoury College Oath for Computing Professionals

Today, I join the ranks of computer scientists worldwide.

I will remember that I remain a member of society, with special obligations to all my fellow human beings.

I will design and build computing systems that enhance the quality of daily life for individuals and for society.

I will protect the dignity of users and others affected by computing systems, respecting the diversity of all cultures and safeguarding against threats to health and safety.

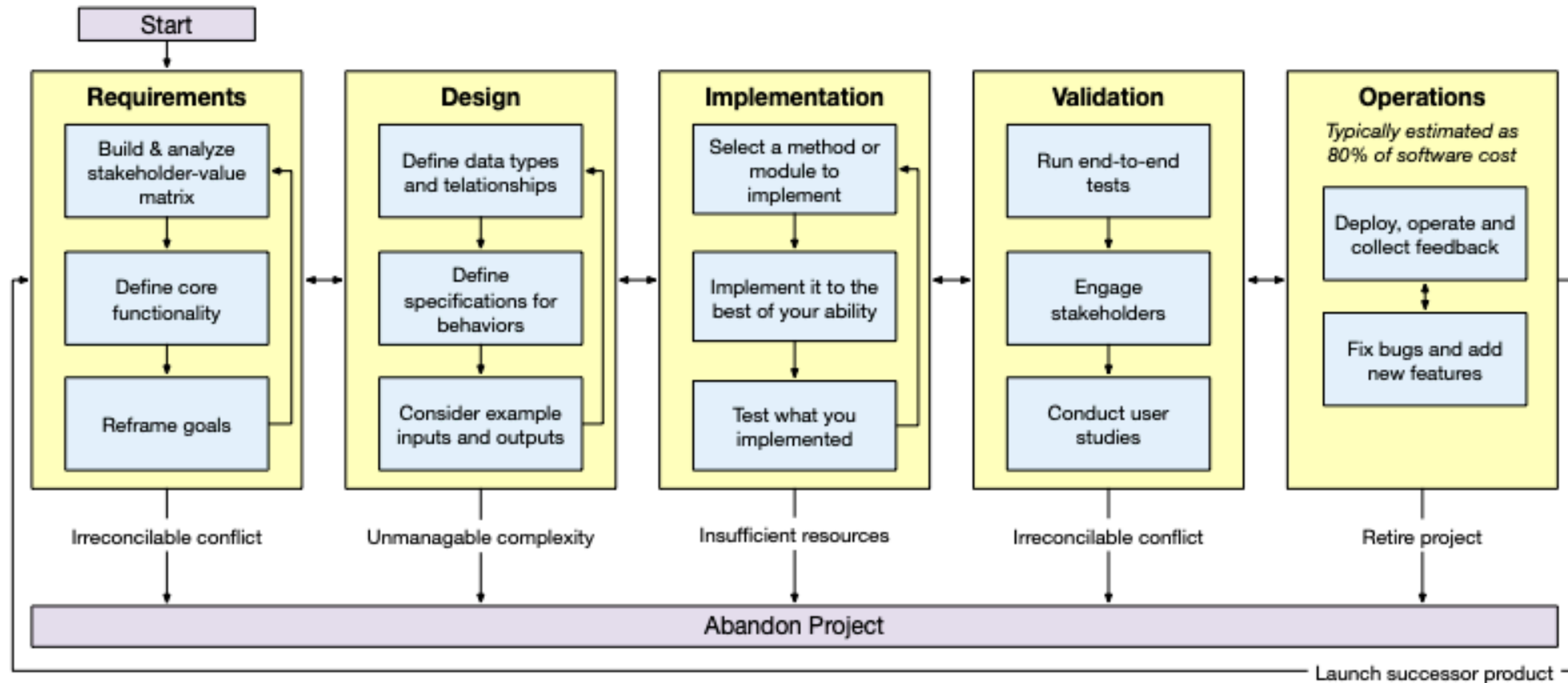
I will respect the privacy and rights of all people and recognize the special role I have in judiciously collecting, storing, and using their information, and creating systems that aim to shape their behavior.

I will work for fair wages, honorably guarding my reputation and my colleagues in our work practices while respecting the intellectual contributions of others.

I will improve the public understanding of computing and its consequences.

May I always act to preserve the finest traditions of my field, and may I long experience the joy of inventing the future through my endeavors.

The Program Design and Implementation Process



VSD happens at the beginning (helps shape the core functionality)

Open-ended poll:

What would happen if we directly jumped to the Implementation phase, skipping the Requirements and Design phases? I.e., what would happen if we started by writing code without first designing the project and planning out its pieces?

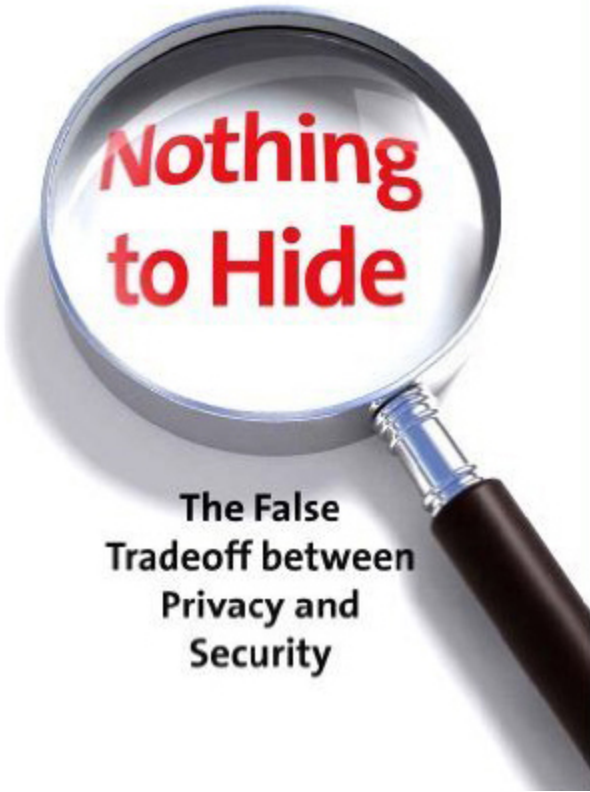
Scenario Discussion 1

Scenario Discussion 2

Why Value-Sensitive Design?

- Cover the vocabulary for moral language to be able to talk with your teammates and advocate for what you think is right
- To debunk the common excuse of "I have nothing to hide"

DANIEL J. SOLOVE



**Nothing
to Hide**

**The False
Tradeoff between
Privacy and
Security**

Example 1: Social media

Social media apps were built to connect communities.

But there are stakeholders with conflicting values:

- The **social media platform** company makes money when more users look at their content (and advertisements) for longer. **Financial values** -> optimize platform to keep users looking at the screen for longer
- Conflicts with **users'** value of **mental health**

Example 2: Prioritizing viewing

Two products:

- reCAPTCHA: verify that users are human
- Duolingo: teach new languages

Conflicting stakeholder values:

- Both use data generated by users to train AI models -> unpaid labor
- Duolingo's financial values also compel them to make users view their app more frequently, leading them to prioritize viewing rather than learning.

Steps to demonstrate Value-Sensitive Design in assignments

1. **Identify the stakeholders:** identify people who are affected by the software in any way
2. **Identify the values:** identify the values at stake for those stakeholders when considering the software
3. **Fill out a stakeholder-value matrix:** create a table, where the columns are the values, and the rows are the stakeholders, and write in how each stakeholder's value relates to the software
4. **Identify and analyze conflicts in the matrix:** for each conflict in the matrix, identify whether and how to address it

VSD Case study: Finder, a dating app

Consider “Finder,” a fictitious dating app (such as [Hinge](#) or [Grindr](#))

Let's:

1. Identify the stakeholders
2. Identify the values
3. Fill out a stakeholder-value matrix
4. Identify and analyze conflicts in the matrix

Step 1: Identify the stakeholders

Users:

- nonbinary people
- transgender people
- people who date people of all genders
- people whose sexual orientation is criminalized in their geographic location
- people who are not legally allowed to be in their geographic location altogether
- sex workers
- survivors of intimate partner violence
- people who are blind
- people who don't speak English
- people whose **names** have apostrophes in them

Step 1: Identify the stakeholders

- **content moderators** (subject to viewing harmful content)
- **software engineers** and all the other **employees** of the company which is building Finder
- **advertisers** (if platform has ads)
- **other customers** of the advertising companies
- **actors and influencers** doing the advertising
- **Governments** and **regulators** (following laws and regulations, protecting the public)
- **Anyone who dates** (dating apps have fundamentally changed the way we date)
- relationships formed with Finder will impact many more stakeholders
- **Stakeholders who will get created** over time

Step 2: Identify the values

- **Privacy:** hide their identity from the government, hide their location from an abusive person
- **Safety** separate value or a subset of privacy
- **Fairness:** valued if we prioritize it. We only want to take into account users' personal preferences, and nothing else, when recommending people as potential dating partners
- **Financial cost** to advertisers and users (depending on Finder's financial model)
- **Reputation**
- **Laws and regulations**

Step 2: Identify the values

- **Accessibility:** different things to different users
 - Nonbinary people may not have access to Finder if we don't include them as a gender option
 - People who are blind or who don't speak English may need us to design Finder so that screen readers and translation apps can work
 - Sex workers are **often unfairly banned** from dating apps, even if they are not doing sex work or soliciting clients through the dating app
 - Need to support a large variety of users' names so that they can create profiles

Step 3: Fill out a stakeholder-value matrix

Matrix:

- columns represent the values
- rows represent the stakeholders
- cells describe how each stakeholder interacts with each value

	Privacy	Accessibility	Fairness	Financial cost	Reputation	
People with criminalized sexual orientations	Hide sexual orientation from government	Able to search profiles of appropriate genders	Should be recommended fairly	Keep costs reasonable		G tc
Survivors of intimate partner violence	Keep location hidden	2FA etc. to prevent account takeover				H al re ac al

	Privacy	Accessibility	Fairness	Financial cost	Reputation	Laws and regulations	to 1 co
People who are blind		Screen readers					
People who are not allowed in their location	Keep location hidden from gov't	Set up profile without gov't ID					

	Privacy	Accessibility	Fairness	Financial cost	Reputation	Laws and regulation
People who don't speak English	Avoid surveillance	Able to use translation apps				
People whose names have apostrophes		Able to set up a profile				

	Privacy	Accessibility	Fairness	Financial cost	Reputation	Law regul
Content moderators	Avoid leaking identity	Access to what they need, nothing more	Training to ensure consistency	Compensated for seeing harmful content	Uphold company reputation	Laws speed personal data
Advertisers	Hide financial details, etc.	Want users' contact information		Reach more audiences for less money	Be seen on good platforms	Follow regul

Step 4: Identify and analyze conflicts in the matrix

- Advertisers want to keep financial costs low, but that may push the financial cost onto the users or the company building Finder
 - Must choose one over the other
- Users want to keep identities / locations private from the gov't, but gov't may want access to that info to enforce laws / protect the public
 - Keep all information private until regulators subpoena
 - Only provide specific info to gov't (not location / sexual orientation)

Step 4: Identify and analyze conflicts in the matrix

- Fairness or accessibility can sometimes unfortunately require risking Finder's reputation, depending on the audience giving reputation-related opinions / how strongly they feel that others don't belong
 - Personally, I think fairness would increase reputation among the right audiences
- Ensuring fairness for users can require collecting personal information, which can then compromise privacy
 - Find a balance of which info to collect
 - Don't share that info (only use it to ensure fairness)
- Criminalized populations want to exist despite being targeted by gov't
 - Don't share info with gov't unless absolutely legally required

Bias and Unfairness (throughout the process)

- We will not solve society's issues with a VSD matrix
 - That's a more challenging and continuous process
- But we can mitigate bias and unfairness in our technology
 - More next time!

Polls: Your Case Study: Algorithmic hiring

"Shamazon" (a fictitious company) is looking to hire software engineers, and you have been tasked with designing a tool to filter the submitted resumes and select the ideal candidates for hire.

- Who are the stakeholders?
- What are the values?
- What is in the stakeholder-value matrix?
- What are the conflicts in the matrix? And how can we mitigate them?
- Where are bias and unfairness entering the product? And how can we mitigate them?

Poll:

- 1. What is your main takeaway from today?**
- 2. What would you like to revisit next time?**